



PROJECT AND TEAM INFORMATION

Project Title

(Try to choose a catchy title. Max 20 words).

Smart Courier Delivery & Route Optimization System

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PHASE-II PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract (2 pts)

(Brief restatement of your project's main goal. Max 300 words).

Our project is a C++ based courier delivery system that helps manage parcels and plan efficient delivery routes. Instead of finding only one shortest path, the system will handle several parcels, different vehicles and delivery priorities. The road network will be stored as a weighted graph. Dijkstra algorithm will find the shortest path between locations. A priority queue will help handle urgent parcels first. For multiple deliveries, the Clarke-Wright Savings Algorithm will create an initial set of routes, while 2-Opt will be used to improve those routes. Vehicle capacity will also be checked before a route is finalized. The aim is to build a practical project that combines Data Structures, algorithms and C++ OOP concepts in one system.

Updated Project Approach and Architecture (2 pts)

(Describe your current approach, including system design, communication protocols, libraries used, etc. Max 300 words).

We will build the project as a menu-driven C++ application using classes such as Parcel, Vehicle, Location, Graph and Route Optimizer. The road network will be represented as a weighted graph. Parcels will contain details such as ID, destination, weight and priority. A priority queue will manage urgent deliveries. Dijkstra algorithm will calculate shortest road paths. The Clarke-Wright Savings Algorithm will combine delivery points into practical vehicle routes while checking vehicle capacity. The 2-Opt method will then check whether a generated route can be shortened. We will use the C++ Standard Template Library for vectors, queues and priority queues, along with basic file handling for saving project data. Each module will be tested separately before the complete system is integrated.

Tasks Completed (7 pts)

(Describe the main tasks that have been assigned and already completed. Max 250 words).

Task Completed	Team Member
Requirement analysis and project scope finalized.	Arpit Rai
Graph, parcel and vehicle data-structure design.	Shreshth Vardhan
Main algorithm flow finalized: Dijkstra, Clarke-Wright, 2-Opt and priority queue.	Both
Initial OOP class structure and menu flow planned.	Both

Challenges/Roadblocks (7 pts)

(Describe the challenges that you have faced or are facing so far and how you plan to solve them. Max 300 words).

The main challenge is combining several algorithms without making the program unnecessarily complicated. Dijkstra works on individual shortest paths, while Clarke-Wright combines several deliveries into routes, so the two parts need to be connected carefully. Another challenge is keeping vehicle capacity within limits while routes are changed during optimization. We also need to make sure that 2-Opt improves a route without breaking the delivery order or capacity condition. To handle these issues, we will keep each algorithm in a separate C++ module and test it independently before integration. We will start with small sample graphs and a few parcels, then increase the test cases gradually.

Tasks Pending (7 pts)

(Describe the main tasks that you still need to complete. Max 250 words).

Task Pending	Team Member (to complete the task)
Implement the graph and Dijkstra module.	Shreshth Vardhan
Implement parcel, vehicle and priority-queue modules.	Arpit Rai
Implement Clarke-Wright route construction.	Both
Add 2-Opt route improvement and capacity checks.	Both
Integrate modules, test, debug and prepare final demo.	Both

Project Outcome/Deliverables (2 pts)

(Describe what are the key outcomes / deliverables of the project. Max 200 words).

The expected deliverable is a working C++ prototype that can store parcel information, manage delivery priorities and create routes for available vehicles. It will use a weighted graph for the road network, Dijkstra for shortest paths, Clarke-Wright for creating multi-stop routes and 2-Opt for improving them. The system will also check vehicle capacity and display the final route, distance, parcel list and vehicle details. The project will demonstrate the practical use of Data Structures, algorithms and OOP concepts in one application.

Progress Overview (2 pts)

(Summarize how much of the project is done, what's behind schedule, what's ahead of schedule. Max 200 words.)

At the current stage, the project scope, main features and algorithmic flow have been finalized. The team has also planned the main classes, data structures and menu flow. The next major part is implementation: building the graph, adding parcel and vehicle management, and coding the four main algorithmic components. Testing and route comparison will follow after the modules are integrated. The project will not use live GPS or traffic data in the current version. Our immediate focus is to complete a stable prototype and then improve its route optimization and user experience.

Codebase Information (2 pts)

(Repository link, branch, and information about important commits.)

The project code is being organized as a C++ application with separate modules for parcels, vehicles, graph operations and route optimization. A public repository link and branch name have not been finalized yet. The first commits will cover the basic project structure, data models and graph implementation; later commits will add Dijkstra, Clarke-Wright, 2-Opt, testing and integration.

Testing and Validation Status (2 pts)

(Provide information about any tests conducted)

Test Type	Status (Pass/Fail)	Notes
Graph / Dijkstra Weighted graphs.	Pending	Test shortest paths on small
Priority queue first	Pending	Check urgent parcels are selected
Route optimization	Pending	Compare routes before and after 2-opt

Deliverables Progress (2 pts)

(Summarize the current status of all key project deliverables mentioned earlier. Indicate whether each deliverable is completed, in progress, or pending.)

Project structure & team planning ?
Completed.
Graph & data-structure design ?
Completed.
Core C++ implementation ?
In progress.
Dijkstra implementation ?
Pending.
Clarke-Wright & 2-Opt ?
Pending.
Vehicle-capacitycheck?
Pending.
Testing & debugging ?
Final Documentation & Demonstration?
Pending.